

ValveTrack

How the Job Card System Works & How to Present It

A plain-language guide to the digital Job Card — the single record that carries a valve from order receipt through welding, heat-treatment, inspection, dispatch and payment — and a step-by-step script for demonstrating it to your client.

Live system

<https://trace-forge-xi.vercel.app> · Sample Job Card: RRE-2026-0019

PART 1

What the system is, in one minute

ValveTrack replaces the paper "Job Card" traveller that follows each valve around the shop floor. Everything that used to be written by hand — welding parameters, heat-treatment charts, inspection readings, sign-offs, dispatch and payment — is now captured in one digital record per job. That record **cannot advance to the next stage until the required document or approval for the current stage exists**. Those rules are enforced by the database itself, so a step can never be skipped by accident, even by mistake on the screen.

One job = one Job Card = one page. After you create a job, the entire rest of the workflow happens on that job's detail page, as sections stacked top to bottom. You rarely leave it.

PART 2

Who does what — the 7 roles

Every person logs in with a role. The role decides which menu items and actions they see. One password for the demo accounts: **Test@123**.

Role	What they can do
Admin	Full control — overrides, portal users, settings, all gates
QA	Approvals, master data, all reports, dossiers
Engineer	Process execution, PWHT, reports
Operator	Job cards, basic data entry, operation status
Accounts	Dispatch, invoicing, payments, alerts
Management	Read-only oversight + dashboard
Customer	Portal — sees only its own company's jobs

Customers log in at a separate address (</portal>) and see **only their own company's jobs** — never another client's. This isolation is enforced by the database, not by hiding things on screen.

The 13-stage life of a Job Card

A job moves down this list in order. The stages marked **GATE** are the points where the system blocks progress until a condition is met — these are the heart of "no document, no progress."

#	Stage (what staff see)	Meaning / rule
1	Received	Job Card created; auto-numbered (RRE-YYYY-NNNN).
2	Awaiting WPS	Welding Procedure Spec must be attached.
3	WPS Under Review	QA/Admin reviews the WPS.
4	WPS Approved	GATE: an approved WPS is required to proceed.
5	Production Assigned	Operations routed to machines/operators.
6	In Production	Each operation run in sequence (no skipping).
7	Production Complete	GATE: every routed operation done or skipped.
8	Inspection In Progress	NDE, PMI, Dimension, Air-test recorded.
9	Inspection Cleared	GATE: an approved inspection report exists.
10	Ready for Dispatch	GATE: approved WPS + (if req'd) approved PWHT.
11	Dispatched	DC number, vehicle, delivery recorded.
12	Invoicing	Accounts records invoice + payment.
13	Completed	GATE: dispatch exists + payment received.

The 5 hard gates (database-enforced)

- **WPS Approved** — no production without an approved welding procedure.
- **Production Complete** — every routed operation must be completed or explicitly skipped (admin-authorized).
- **Inspection Cleared** — at least one approved inspection report (PMI, Dimension or Overlay).
- **Ready for Dispatch** — approved WPS and, if the WPS requires it, an approved PWHT heat-treatment run.
- **Completed** — a dispatch record exists and payment has been received.

PART 4

Production routing & the no-skip rule

When a job reaches production, its work is broken into an ordered **routing** of operations — for a typical body: Pre-Machining ' Welding ' Final Machining ' Milling ' Slitting ' Deburring. Each operation carries a machine, an operator, and planned / completed / rejected quantities.

Sequential enforcement (Phase 3)

- An operation **cannot start or complete until every earlier step is completed or skipped**.
- If a non-admin tries to jump ahead, they get a clear message naming the blocking step(s) — e.g. "Complete earlier steps first: pre_machining (#1)".
- Only an **Admin** may force an out-of-order start, or mark a step **Skipped** (N/A). Either action is stamped with **who** did it and **why**, and shows an amber **Override** badge on the card.

PART 5

Due dates & aging (at-a-glance urgency)

Each job can carry a production **Due Date**. Lists and the dashboard colour-code urgency automatically:

🟢On schedule 🟡Due soon (within 3 days) 🟠Overdue 🟣Done (off the floor)

- **Aging** = days since the job was created, shown on the list so long-running jobs stand out.
- The dashboard has an **Overdue** tile counting jobs past their due date that are still on the floor. Jobs already dispatched/closed never raise a false alarm.
- Existing jobs can be back-filled with a due date from the inline editor on the job's Tracking card.

PART 6

The generated Job Card PDF

From any job you can **Generate PDF** — a faithful digital copy of the paper traveller, now including a **Due Date**, a full **Operations / Routing** table (every step with machine, operator, quantities and status), PWHT, NDE + chemicals, air-test, the machining grid, dispatch and sign-offs. Welding parameter details only appear for welding operations. See the attached **JobCard_Sample_RRE-2026-0019.pdf**.

How to present it — a 10-minute demo script

Open <https://trace-forge-xi.vercel.app>. Use the coloured quick-login buttons so you never type a password. Follow the sample job **RRE-2026-0019**, which is fully populated end-to-end.

● Log in as Admin

Click the Admin quick-login. Land on the Dashboard — point out the 5 live stat tiles and the red Overdue tile. Say: 'every number here is live from the database.'

● Open the Dashboard's story

Show Recent Job Cards and the Blocked / Attention list. Explain the colour dots = due-date urgency.

● Open Job Card RRE-2026-0019

Sidebar ' Job Cards ' open the sample. Say: 'this one page is the entire job.' Scroll slowly through the stacked sections.

● Show the gates

Point at WPS (approved) ' Operations routing (6 steps, all completed, machines + qty) ' Inspection reports ' Dispatch ' Accounts. Emphasise: the job only reached the end because each gate was satisfied.

● Demonstrate the no-skip rule

On a fresh job (or describe it): try to start step 2 before step 1 — the system refuses and names the blocker. Only Admin can override or skip, and it's recorded.

● Generate the PDF

Click Generate PDF. Open it. Point out the Operations/Routing table and Due Date — 'the paper traveller, produced automatically.'

● Switch roles

Log out, log in as Operator — show the sidebar is smaller (no Master Data, no approvals). Then Accounts — show they focus on dispatch/payment. 'Everyone sees only what their job needs.'

● Show the Customer Portal

Go to /portal as a client login — they see only their own jobs, read-only. 'Your customer gets transparency without seeing anyone else's work.'

PART 8

How Accounts works

When a job is dispatched, it moves to **Invoicing**. The Accounts team records, on the job's Accounts section:

- Invoice number & date, invoice amount, and the tax/GST breakdown.
- Payment status and the date payment was received.
- **Close rule:** a job can only be marked **Completed** when a dispatch record exists **and** payment is received (an Admin can override in exceptional cases, and it is logged).

A separate **Accounts_How_It_Works.pdf** walkthrough is available for that team.

PART 9

Demo logins

Role	Email	Password
Admin	admin@raghaveng.com	Test@123
QA	qa@raghaveng.com	Test@123
Engineer	engineer@raghaveng.com	Test@123
Operator	operator@raghaveng.com	Test@123
Accounts	accounts@raghaveng.com	Test@123
Management	management@raghaveng.com	Test@123

PART 10

Audit snapshot (2026-07-09)

- **Type-check** (strict): clean. **Unit tests:** 63/63 pass. **Production build:** green.
- **Security — anonymous access:** blocked on all core tables (job cards, clients, profiles, dispatches, reports).
- **All 22 core tables present** with live data; the unrelated legacy tables have been removed.
- **Workflow gates verified live** — the sample job passes every gate; blockers report correctly when a condition is unmet.
- **Sequence trigger & due-date/overdue logic** verified against live data (3 overdue jobs correctly flagged).
- **Backward compatibility:** job cards created before this update still open and generate PDFs.

Note: the Supabase advisory scan (security/performance linter) could not run in this session because the management connector was offline; it should be re-run once reconnected. It does not affect the functional results above, which were verified directly against the live database.